

ZOO 2222: Vertebrate Diversity (2C)

Lecture 1 & 2: Chordate Characteristics and Protochordates

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Introductory Lecture Questions

1. Which of the following is a defining characteristic shared by all chordates at some point during their development?

- A. Bony vertebral column
- B. Jaws
- C. Notochord (Correct Answer)**
- D. Paired appendages

2. Which of the following vertebrate groups is entirely ectothermic and lays eggs in water during reproduction?

- A. Amphibians (Correct Answer)**
- B. Reptiles
- C. Birds
- D. Mammals

Lecture Objectives

- Define the major characteristics of chordates.
- Identify and describe the key features that distinguish chordates from other animal groups.
- Understand the diversity and classification of chordates.

1. Introduction to Phylum Chordata

The Phylum Chordata represents the most advanced phylum of the animal kingdom. The name is derived from the Greek words 'noton' (meaning back) and 'chorda' (meaning cord). By definition, chordates are organisms that possess a notochord during at least some part of their developmental lifecycle.

Compared to lower invertebrates, chordates exhibit a highly complex anatomy. Key advancements are found particularly in their well-developed nervous and circulatory systems. This diverse phylum includes a wide variety of animals, ranging from primitive marine organisms (protochordates) to familiar vertebrate classes: fishes, amphibians, reptiles, birds, and mammals.

2. The Five Hallmark Characteristics of Chordates

At some stage of their development, all members of Phylum Chordata share five distinctive diagnostic features. These structures are crucial for understanding their body plans, locomotion, and evolutionary lineages:

A. Notochord

The notochord is a flexible, rod-like structure that extends the length of the body, located dorsal to the gut (alimentary canal) but ventral to the central nervous system. It serves as the primary longitudinal axis of support for the body.

- **Embryonic Origin:** Derived from the embryonic mesoderm.
- **Structure:** Composed of fluid-filled cells enclosed by a robust, fibrous sheath and an outer elastic sheath, which provides both stiffness and lateral flexibility.
- **Function:** It acts as the main skeletal axis for muscle attachment. During lateral movement, it stiffens the body and prevents telescoping, facilitating efficient locomotion.
- **Evolutionary Fate:** In primitive chordates, the notochord persists throughout their entire life. In most advanced chordates (vertebrates), it is progressively replaced by a bony or cartilaginous vertebral column during development.

B. Dorsal Tubular Nerve Cord

Unlike the nervous systems found in many invertebrate groups, the chordate nerve cord is a single, hollow, fluid-filled tube of nervous tissue running the length of the animal. It is positioned dorsal to the notochord and the alimentary canal.

- **Key Contrast:** While invertebrates characteristically possess ventral, paired solid nerve cords with segmented ganglia, chordates possess a single, dorsal hollow nerve cord.
- **Cephalization:** The anterior end of this nerve cord typically enlarges and specializes to form the brain.
- **Lifespan Presence:** This structure is present in chordates throughout both embryonic and adult life.

C. Pharyngeal Gill Slits

Pharyngeal slits are paired, lateral clefts leading from the cavity of the pharynx to the exterior environment.

- **Evolutionary Origin:** These structures originally evolved as a specialized filter-feeding apparatus, allowing ancient chordates to filter microscopic food particles from water pumped through the mouth.
- **In Aquatic Vertebrates:** They persist throughout life in fishes and a few tailed amphibians, where they are highly vascularized and modified to function as respiratory gills.
- **In Terrestrial & Higher Vertebrates:** Found only during embryonic development in a highly transient, non-functional state (often forming pharyngeal pouches), later giving rise to structures like middle ear cavities, tonsils, and glands.

D. Endostyle or Thyroid Gland

The endostyle is a glandular groove situated on the floor of the pharynx. It consists of both ciliated and glandular secretory cells that produce mucus designed to trap food particles during filter feeding.

- **Iodine Trap Function:** Certain specialized cells within the endostyle actively concentrate and trap iodine from water.
- **Precursor to Thyroid:** Because of its iodine-binding capabilities, the endostyle serves as the direct evolutionary precursor to the thyroid gland of higher vertebrates. In more derived chordates, it metamorphoses into the thyroid, producing iodine-containing hormones (like thyroxine) that regulate metabolic activities.

E. Post-Anal Tail

The post-anal tail is a muscular structure extending posteriorly beyond the anus.

- **Anatomy:** It is supported by the notochord, the hollow nerve cord, and segmentally arranged muscle blocks called myotomes, all extending into the tail.
- **Functional Diversity:** In aquatic larval forms and fishes, it is primarily used for underwater propulsion and swimming. In terrestrial chordates, it is adapted for leverage, balance, and signaling, rather than active locomotion.
- **In Humans:** The tail is highly reduced and vestigial, represented only by the coccyx (tailbone).

F. Additional Anatomical Features of Chordates

Beyond the five hallmark characteristics, chordates also exhibit the following standard body plans:

1. **Blocks of Muscle (Metameric Musculature):** Muscle segments (myotomes) are arranged on either side of the body, surrounding the spinal cord and notochord to produce coordinated lateral bending movements.
2. **Triploblastic Coelomates:** Body structures are derived from three distinct germ layers (ectoderm, mesoderm, and endoderm) and contain a well-developed, true coelom (body cavity partitioned during embryogenesis).
3. **Bilateral Symmetry:** The body displays clear bilateral symmetry (left and right halves are mirror images along a central sagittal plane).
4. **Organ Systems & Physiology:** Distinct anatomical systems exist including: limbs (four appendages typically in tetrapods), an internal bony or cartilaginous endoskeleton with a backbone, an advanced digestive system, respiration through lungs or gills, excretion via kidneys, a ventral chambered heart, and a closed vascular circulatory system.

3. Taxonomy and Relationships of Chordates

Traditionally, Phylum Chordata is structured into four distinct subphyla, categorizing both invertebrate (protochordate) lineages and the lineage of backboned animals (Vertebrata):

Subphylum	Common Representation	Key Diagnostic Features
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Hemichordata	Acorn worms (Enteropneusta) & Pterobranchs	Marine worms with body split into proboscis, collar, trunk; primitive stomochord; glomerulus for excretion.
Urochordata (Tunicata)	Tunicates (Sea squirts, Salps, Larvaceans)	Sessile or planktonic; notochord and post-anal tail present in larval stages; enclosed in a tough tunic.
Cephalochordata	Lancelets (Amphioxus / Branchiostoma)	Sessile sand-dwellers; clearly show all 5 chordate hallmark traits throughout their entire lives.
Vertebrata (Craniata)	Fish, Amphibians, Reptiles, Birds, Mammals	Notochord replaced by a vertebral column (backbone); highly developed cephalization (brain and cranium).

The Protochordates

Subphyla Hemichordata, Urochordata, and Cephalochordata collectively comprise the 'Protochordates' (primitive chordates). These organisms occupy a fundamental evolutionary bridge between major invertebrate phyla and the vertebrates. They are mostly marine filter feeders and provide critical clues regarding the ancestral state of early chordate body plans.

4. Detailed Analysis of Invertebrate Deuterostomes

A. Subphylum Hemichordata

Hemichordates represent a fascinating group of marine deuterostomes that share several developmental and morphological traits with true chordates, though they lack a true notochord. Instead, they possess a short, rudimentary structure in the collar region known as the stomochord.

General Characteristics:

- **Body Division:** The body is divided into three functional parts: an anterior proboscis, a middle collar, and a posterior trunk.
- **Symmetry & Layers:** Bilaterally symmetrical with three distinct germ layers (triploblastic).
- **Coelom:** A true coelom is present, split into three compartments corresponding to the three body regions (protoel, mesocoel, and metacoel).
- **Digestive System:** Complete U-shaped or straight digestive tract ending in a functional anus.
- **Nervous System:** Diffuse and variable; consists of a dorsal nerve cord (in the collar region) alongside a smaller ventral nerve cord.
- **Excretory System:** Employs a specialized glomerulus located within the proboscis to filter

nitrogenous waste from the blood.

- **Reproduction & Development:** Usually sexual. Shows indirect development via a free-swimming, transparent, and planktonic Tornaria larva, which is structurally similar to echinoderm larvae.

Major Classes of Hemichordates:

1. **Class Enteropneusta (Acorn Worms):** Solitary, slow-burrowing, worm-like animals found in shallow marine waters where they live in U-shaped mud burrows. They utilize a mucus-coated, ciliated proboscis for deposit or filter feeding, ingesting sand/sediment and excreting mud castings on the seabed (e.g., *Balanoglossus*).
2. **Class Pterobranchia:** Tiny, marine, colonial animals (zooids are < 1 mm) that build and inhabit branching tubes, forming moss-like colonies. They lack a dorsal nerve cord and possess only a single pair of gill slits. Their collar is modified into branched, mucus-secreting tentacles that capture suspended food particles.

B. Subphylum Cephalochordata (Lancelets)

Cephalochordates are small (2-3 inches long), fish-like marine organisms commonly called lancelets or *Amphioxus* (meaning 'sharp at both ends', represented by the genus *Branchiostoma*). They are of profound scientific importance because they cleanly exhibit all five hallmark chordate characteristics throughout their entire adult lives.

- **Habitat & Lifestyle:** Found in sandy seashores of temperate zones, where they live semi-buried in the sand.
- **Feeding:** Ciliary filter-feeders that generate a directed water current into their mouth using cilia lining their gill bars.
- **Cephalization:** They exhibit very minimal cephalization (rudimentary head region, without a true brain or cranium).

C. Subphylum Urochordata (Tunicata / Sea Squirts)

Urochordates are marine filter feeders protected by a tough, non-living, gelatinous outer sheath called a tunic. They illustrate dramatic metamorphosis: while their free-swimming, tadpole-like larvae clearly display a notochord, dorsal hollow nerve cord, and post-anal tail, these structures are completely lost when the larvae metamorphose into highly specialized, sessile or planktonic adults.

Major Classes of Urochordates:

1. **Class Ascidiacea (Sea Squirts):** Sessile, solitary, or colonial bottom-dwellers. They possess two openings (incurrent and excurrent siphons) to draw water in and pump it through a spacious, perforated pharynx to trap organic food particles. (e.g., *Ciona*, the tube sea squirt).
2. **Class Thaliacea:** Small, barrel-shaped, entirely planktonic organisms with incurrent and excurrent siphons at opposite ends of their body. They pump water for filter-feeding and jet-propulsion (e.g., *Salpa*, *Doliolum*).
3. **Class Appendicularia (Larvacea):** Planktonic, transparent organisms that retain larval traits—including a tail containing a notochord—throughout their entire adult life. They construct and live inside a delicate, gelatinous 'house' that acts as an advanced water-filtration system.

5. Modern Evolutionary Phylogeny of Deuterostomes

Recent advancements in molecular biology, genomics, and evolutionary developmental biology (evo-devo) have revolutionized our understanding of deuterostome relationships. Historically, scientists relied solely on adult morphology, which grouped hemichordates closely with chordates due to shared features like pharyngeal gill slits.

However, modern DNA and protein sequence analysis demonstrates that deuterostomes are split into two major monophyletic clades:

- Clade 1: Ambulacraria – Grouping Echinodermata and Hemichordata. This close relationship is strongly supported by shared larval forms (tornaria and bipinnaria larvae), similarities in coelomic organization, and shared genetic markers.

- Clade 2: Chordata – Consisting of Cephalochordata, Urochordata, and Vertebrata.

Cephalochordates diverged earliest, while Urochordata and Vertebrata are classified as sister groups under the clade Olfactores.

Evolutionary Case Study: Molecular vs. Morphological Evidence

A molecular phylogenetic study demonstrates that hemichordates share significantly higher genetic similarity with echinoderms than with chordates, despite possessing typical chordate features like pharyngeal gill slits. Let us examine three critical evolutionary questions:

1. How can molecular evidence override morphological similarity in phylogenetics?

Morphological traits can be highly misleading. Similar structures can emerge independently due to convergent evolution, or can simply represent retained ancestral traits (plesiomorphies) that do not indicate a close evolutionary relationship. Conversely, molecular data (DNA/RNA/proteins) provides a massive, independent dataset of nucleotide positions. This is highly heritable, less prone to environmental adaptation bias, and allows researchers to reconstruct highly accurate and statistically robust evolutionary trees.

2. What is the significance of the clade Ambulacraria?

The Ambulacraria clade (Echinoderms + Hemichordates) forms a key monophyletic group within Deuterostomia. It proves that hemichordates are sister to echinoderms, refuting earlier models that grouped hemichordates directly within the chordate phylum. This reveals that chordates arose as a separate lineage from an early deuterostome ancestor.

3. Why are hemichordates and echinoderms grouped together despite having vastly different adult body forms?

Despite adult differences (bilateral, worm-like hemichordates vs. pentaradial echinoderms), they share identical deuterostome embryonic pathways, highly identical larval forms (tornaria vs. bipinnaria), and robust genomic markers. Their adult differences are a classic example of divergent evolution, where echinoderms evolved secondary radial symmetry, and hemichordates retained ancestral bilateral symmetry.